

SANSKRIT CHARACTER RECOGNITION AND DIGITIZATION USING SOFT COMPUTING

By

Jadeja Pratikshaba S.
(180410754003)

Guided by:

Er. Saurabh M. Patel
Assistant Professor of E&C department

A Thesis Submitted to
Gujarat Technological University in Partial Fulfillment of the requirements for
the Masters of Engineering Degree in Embedded System

June-2020



Electronics & Communication Department,
Sardar Vallabhbhai Patel Institute of Technology,
P.O.: Vasad, Dist.:Anand

CERTIFICATE

This is to certify that research work embodied in this thesis entitled “SANSKRIT CHARACTER RECOGNITION AND DIGITIZATION USING SOFT COMPUTING” was carried out by **Jadeja Pratikshaba S. (180410754003)** at **Sardar Vallabhbhai Patel Institute of Technology (College Code: 41)** for partial fulfillment of Master of Engineering degree in **Embedded System** to be awarded by Gujarat Technological University. This research work has been carried out under my guidance and supervision and it is up to my satisfaction.

Date:

Place: SVIT, Vasad

Guide

Mr.S.M.Patel

Assistant Professor
E&C Department, SVIT

Principal

Dr. S. D. Toliwal
SVIT-Vasad

Seal of Institute

COMPLIANCE CERTIFICATE

This is to certify that research work embodied in this thesis entitled “SANSKRIT CHARACTER RECOGNITION AND DIGITIZATION USING SOFT COMPUTING” was carried out by **Jadeja Pratikshaba S. (180410754003)** at Sardar Vallabhbhai Patel Institute of Technology (College Code: 41) for partial fulfilment of Master of Engineering degree to be awarded by Gujarat Technological University. She has complied to the comments given by the Dissertation phase – I as well as Internal Review-II Reviewer to my satisfaction.

Date:

Place: SVIT, Vasad

Jadeja Pratikshaba S.
(180410754003)

Dr. K.R.Bhatt
HOD
E&C Department, SVIT

PAPER PUBLICATION CERTIFICATE

This is to certify that research work embodied in this thesis entitled “SANSKRIT CHARACTER RECOGNITION AND DIGITIZATION USING SOFT COMPUTING” was carried out by **Jadeja Pratikshaba S. (180410754003)** at **Sardar Vallabhbhai Patel Institute of Technology (College Code: 41)** for partial fulfillment of Master of Engineering degree to be awarded by Gujarat Technological University has published articles **A Review Character Recognition and Digitization of Sanskrit Characters** in **Journal of The Gujarat Research Society(JGRS)** Volume 21, Issue 13, **December 2019** and **Machine learning and Deep Learning Approaches for Sanskrit Character Recognition** in the **Journal of Xidian University** volume 14, Issue 05 during/on **May 2020**.

Date:

Place: SVIT, Vasad

Jadeja Pratikshaba S.
(180410754003)

Mr. S.M. Patel
Assistant Professor
E&C Department, SVIT

Dr. S.D.Toliwal
Principal, SVIT

Seal of the Institute

THESIS APPROVAL CERTIFICATE

This is to certify that research work embodied in this thesis entitled “ SANSKRIT CHARACTER RECOGNITION AND DIGITIZATION USING SOFT COMPUTING” was carried out by **Jadeja Pratikshaba S. (180410754003)** at **Sardar Vallabhbhai Patel Institute of Technology (College Code: 41)** is approved for the degree of Master of Engineering with specialization in **Embedded System** by Gujarat Technological University.

Date:

Place:

Examiner's Sign and Name:

()

()

समुचित

ज्ञान

समन्वय

UNDERTAKING ABOUT ORIGINALITY OF WORK

We hereby certify that we are the sole authors of this thesis and that neither any part of this thesis nor the whole of the thesis has been submitted for a degree to any other University or Institution.

We certify that, to the best of our knowledge, the current thesis does not infringe upon anyone's copyright nor violate any proprietary rights and that any ideas, techniques, quotations or any other material from the work of other people included in our thesis, published or otherwise, are fully acknowledged in accordance with the standard referencing practices. Furthermore, to the extent that we have included copyrighted material that surpasses the boundary of fair dealing within the meaning of the Indian Copyright (Amendment) Act 2012, we certify that we have obtained a written permission from the copyright owner(s) to include such material(s) in the current thesis and have included copies of such copyright clearances to our appendix.

We declare that this is a true copy of thesis, including any final revisions, as approved by thesis review committee.

We have checked write up of the present thesis using anti-plagiarism database and it is in allowable limit. Even though later on in case of any complaint pertaining of plagiarism, we are sole responsible for the same and we understand that as per UGC norms, University can even revoke Master of Engineering degree conferred to the student submitting this thesis.

Signature of Student:

Name of Student: Jadeja Pratikshaba S.

Enrolment No: 180410754003

Signature of Guide:

Name of Guide: Mr. S.M.Patel

Assistant Professor, E&C
Department, SVIT

Institute Code: 41

ACKNOWLEDGEMENT

It gives me immense pleasure to thank **Mr. S.M.Patel** (Assi.Prof, E&C Department, SVIT-Vasad) who took great efforts in providing good guidance and was enthusiastic about my work. I learnt from her the skills of writing the report, and also the patience and persistence required to analyze one's own work and address peer review comments. I owe her a great debt of gratitude.

I would like to thank and express my deep and sincere sense of gratitude to **Dr. K.R. Bhatt**(Head, E&C Department, SVIT - Vasad) for providing me encouragement and good facilities to prepare my research. Besides, I am also grateful to all the faculties of E&C Department for providing timely supports and sharing of knowledge.

These acknowledgements would be incomplete without thanking my fellow M.E students for always encouraging each other to work hard.

Finally, I am indebted to my family members, who have encouraged me and provided the moral support throughout till date.

Jadeja Pratikshaba S.
(180410754003)

TABLE OF CONTENTS

CHAPTER	PAGE NO.
TITLE PAGE	I
CERTIFICATE PAGE	II
COMPLIANCE CERTIFICATE	III
PAPER PUBLICATION CERTIFICATE	IV
THESIS APPROVAL CERTIFICATE	V
UNDERTACKING ABOUT ORIGINALITY OF WORK	VI
ACKNOWLEDGEMENT	VII
TABLE OF CONTENTS	VIII
LIST OF FIGURES	XI
ABSTRACT	XIII
CHAPTER1: INTRODUCTION.....	1
1.1 History.....	1
1.2 Motivation.....	2
1.3 Objective.....	3
1.4 Basic System diagram.....	3
Summary.....	3
CHAPTER 2: LITERATURE SURVEY.....	4
2.1 Segmentation Techniques	4
2.1.1 Thresholding.....	4
2.1.2 K-NN.....	4
2.1.3 Pixel cluster Detection Technique.....	4
2.2 Feature Extraction Technique	5
2.2.1 Zoning	5
2.2.2 Projection Profile.....	5
2.2.3 Histogram of Gradients.....	6
2.2.4 The Fuzzy Hough Transform.....	6

2.3 Classification	6
2.3.1 SVM.....	6
2.3.2 ANN.....	7
2.3.3 CNN.....	8
2.3.4 Deep Learning.....	9
2.3.5 Hidden Markov Model.....	10
CHAPTER 3: IMPLEMENTATION PLATFORM AND PROPOSED SYSTEM...	11
3.1 Implementation method.....	11
3.1.1 Anaconda Navigator.....	11
3.1.2 Spyder.....	11
3.2 Proposed System.....	11
3.2.1 Datasets.....	12
3.2.2 Preprocessing.....	12
3.2.3 Segmentation.....	14
3.2.4 Feature Extraction.....	15
3.2.5 Classifiers.....	16
CHAPTER 4: RESULTS.....	19
4.1 Preprocessing on grey scale image.....	19
4.2 Preprocessing on coloured scanned image of document.....	20
4.3 Segmentation.....	23
4.4 Classifier Output.....	24
4.5 Final Output in the form of GUI.....	25
CHAPTER 5: WORK PLAN.....	31
CHAPTER 6: SUMMARY.....	32
REFERENCES.....	33
APPENDIX A. PHOTOCOPY OF REVIEW CARD.....	35
APPENDIX B. ABBREVIATION.....	39
APPENDIX C. COMPLIANCE REPORT OF COMMENTS	40
APPENDIX D. PLAGIARISM REPORT.....	41

LIST OF FIGURE

Sr. NO	NAME	PAGE NO
Figure 1.1:	Basic characters in Devanagari script.....	2
Figure 1.2:	Example of compound characters that are considered unique classes.....	2
Figure 1.3:	Basic block diagram.....	3
Figure 2.1:	SVM classifier.....	7
Figure 2.2:	Architecture of back propogation neural network.....	8
Figure 2.3:	Convolution Neural network.....	9
Figure 2.4:	Deep neural network.....	9
Figure 2.5:	HMM classifier.....	10
Figure 3.1:	Proposed system diagram.....	11
Figure 3.2:	Sample dataset of Sanskrit characters	12
Figure 3.3:	DWT Block Diagram.....	16
Figure 3.4:	Proposed CNN Layers.....	17
Figure 4.1:	Original scanned document	19
Figure 4.2:	Applying median filter on document.....	19
Figure 4.3:	Applying gaussian filter on document	20
Figure 4.4:	Applying bilateral filter on document	20
Figure 4.5:	Applying Average filter on document	20
Figure 4.6:	Original coloured scanned document	21
Figure 4.7:	Applying median filter on document.....	21
Figure 4.8:	Applying gaussian filter on document	22
Figure 4.9:	Applying bilateral filter on document	22
Figure 4.10:	Applying Average filter on document	22
Figure 4.11:	Output after segmentation.....	23
Figure 4.12:	Segmentation on document of different font size.....	23
Figure 4.13:	Model Accuracy Graph.....	23
Figure 4.14:	Model Loss Graph.....	24
Figure 4.15:	Model Graphs as Output in GUI.....	25

Figure 4.16:	Output on clicking on Browse Button.....	26
Figure 4.17:	Segmented and filtered image as output.....	26
Figure 4.18:	Recognizing of letters, words and sentences.....	27
Figure 4.19:	Digitized form of document.....	28
Figure 4.20:	Recognizing of Sanskrit words printed on coloured background.....	29
Figure 4.21:	Recognizing number entity.....	30

LIST OF TABLE

Sr. NO.	NAME	Page No.
Table No.4.1	Time Required to train and test various classifier.....	24
Table No.4.2	Existing Vs Proposed.....	30
Table No.5.1	Work-Plan.....	31

Sanskrit Character Recognition and Digitization using Soft Computing

Submitted By

JADEJA PRATIKSHABA S.

Supervised By

Mr. SAURABH M. PATEL
ASSISTANT PROFESSOR
SARDAR VALLABHBHAI PATEL INSTITUTE OF TECHNOLOGY
VASAD

ABSTRACT

The main objective of any character recognition system is to achieve modification or conversion of any form of text such as printed or scanned text images. Most of the greatest literature works in ancient India were written in Sanskrit. Hence it is necessary to preserve these highly prestigious documents by digitizing them. For digitization of these documents it is necessary that these characters get accurately recognized by the machine for which it needs huge data set as Sanskrit has most number of letters compared to any other languages. This proposed work would not only focus on detection of characters of poorly maintained documents but also work on segmentation of overlapping lines, touching characters in middle and upper zone, consonant segmentation and half character segmentation which would increase the accuracy of the present OCR system for recognition of poorly maintained Sanskrit literature.

CHAPTER 1

Introduction

Most of the greatest literature works to come out of India were written in Sanskrit. It is regarded as the mother of all modern languages. The Vedas are the soul of Indian culture and tradition which were written in Sanskrit even the core thoughts of Buddhism were within the Sanskrit language. Sanskrit is gaining importance in various academic communities due to the presence of ancient scientific and mathematical research work written in this language. Scientists all over the world, are spending increasing amount of time trying to understand these ancient research manuscripts. However, the lack of accurately digitized and tagged versions of Sanskrit manuscripts is a major bottleneck. In addition to this, the poor maintenance and text quality adds to the problem. Hence, it becomes essential to digitize such ancient manuscripts which are not only important for research but, are also an important part of the culture and heritage of India.

1.1 History

Ancient Sanskrit manuscripts are a rich source of knowledge about Science, Mathematics, Hindu mythology, Indian civilization, and culture. It therefore becomes critical that access to these manuscripts is made easy, to share this knowledge with the world and to facilitate further research on this Ancient literature. Sanskrit contains highest number of characters which include 13 vowels and 34 consonants. Apart from the vowels and consonants, there are compound (composite) characters in most of Indian scripts including Devanagari, which are formed by combining two or more basic characters. The shape of a compound (composite) character is usually more complex than its constituent characters.

In order to develop a robust OCR system which can digitize soiled and noisy documents with high accuracy, Meduri Avadesh and Navneet Goyal[1] propose the use of Convolution neural networks (convnets) as opposed to the popular use of SVMs and ANNs, as convnets possess very high learning capacity and the capability to handle high dimensional data such as images. Early efforts in Indian OCRs involve a template based and feature-based approach for letter classification. In the template based approach, each unknown letter or pattern is compared with a standard template pattern and the degree of similarity between the two patterns is used to decide classification. Results of the early OCRs were further improved by using feature-based approaches along with the traditional template based approach. Feature-based

approaches identify unique characteristics of patterns or letters and utilize them for classification.

1.2 Motivation:

Sanskrit is gaining importance in various academic communities due to the presence of ancient scientific and mathematical research work written in this language[1].

अ आ इ ई उ ऊ ए ऐ ओ औ
क ख ग घ ङ
च छ ज झ ञ
ट ठ ड ढ ण
त थ द ध न
प फ ब भ म
य र ल व
श ष स
ह ळ

Figure 1.1 Basic characters in Devanagari script [1]

Scientists all over the world, are spending increasing amount of time trying to understand these ancient research manuscripts. However, the lack of accurately digitized and tagged versions of Sanskrit manuscripts is a major bottleneck. In addition to this, the poor maintenance and text quality adds to the problem. Hence, it becomes essential to digitize such ancient manuscripts which are not only important for research but, are also an important part of the culture and heritage of India. Even though both Hindi and Sanskrit are written in the Devanagari script, it is important to use a Sanskrit OCR instead of a Hindi OCR to digitize Sanskrit text due to the significant difference in complexity between the two languages. Sanskrit text consists of several compound characters which are formed by different combinations of half letter and full letter consonants. Some examples of compound characters are shown in Fig 3 and Fig 4. Since such compound characters are either less frequent or completely absent in Hindi text, Hindi OCRs would not be trained to segment and classify such characters correctly. Subsequently, the Hindi OCRs would display poor results in Sanskrit text.



Fig: 1.2 Examples of compound characters that are considered as unique classes ^[1]

1.3 Objective

To develop an OCR that could accurately recognize and digitize sanskrit documents that are poorly maintained and also have different fonts, have overlapping characters, touching lines and faded letters.

1.4 Basic System diagram:

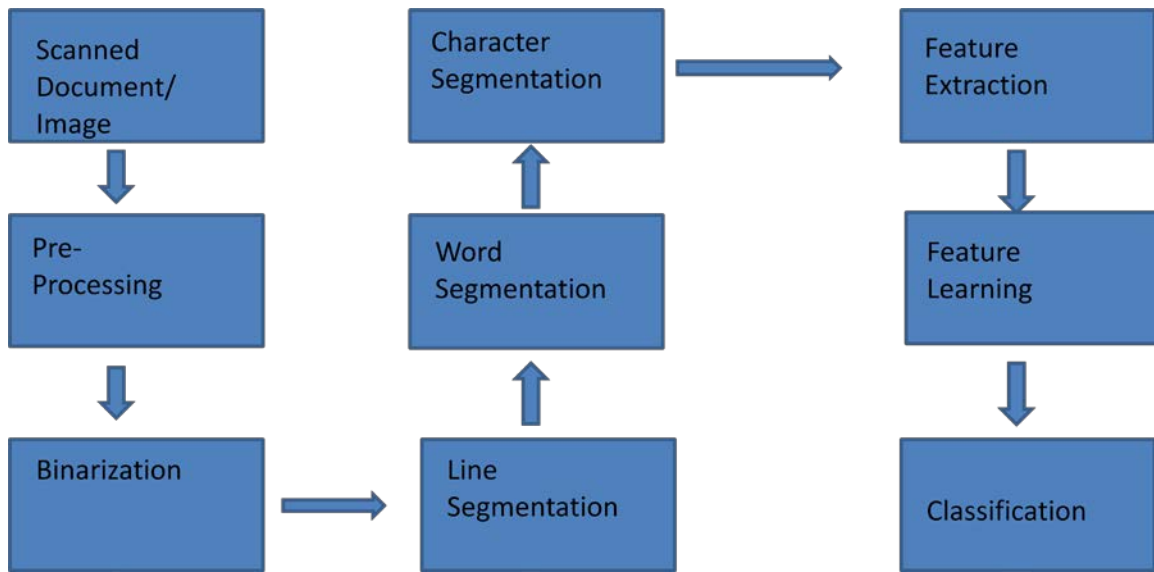


Fig.1.3: basic block diagram

In this block diagram shown in figure 1.3 the flow below is firstly the scanned document (image). Which is then Pre-processed to removes noise from image after removal of noise image is then binarized i.e. converted in black and white from grey scale, followed by line segmentation, word segmentation, character segmentation Feature Extraction and classification.

Summary: In this chapter there are basic introduction of what is an OCR system what are the process required to develop an OCR system and why is it necessary to develop an OCR which could accurately digitize ancient Sanskrit documents.

CHAPTER 2

Literature Survey

Literature survey helped to find out the work done on OCR systems and specially work done for development of Sanskrit OCR by various researchers also what are the challenges faced by them . following are various methodologies which were studied while literature survey on various OCR systems which include various Segmentation methods, various Feature extraction Methods and at last different classifiers for OCR.

Methodology:

2.1 Segmentation:

Segmentation is an important phase of OCR, because it can reach in separation of words, lines or characters . In fact correct recognition based on correct segmentation.

2.1.1 Thresholding:

The processed thresholded images are used to identify and segment letters. Letter segmentation is carried out by identifying the corresponding line and word in the image first. The lines in the text are identified by calculating a row-wise black coloured pixel density. Subsequently, a line is bound to exist between any two local minima as the minimas represent the white spaces between the lines. Following this, the words in the segmented line are identified by calculating the columnwise black pixel intensity. As a result, the segment of the image between any two consecutive local minima contains a word. In an Optical character recognition on heterogen-eous SoC for HD automatic number plate recognition system .[11] thresholding is used for segmentation purpose.

2.1.2 K-NN:

In pattern identification, k-NN is a non-parametric method used for classification and regression. In both circumstances, the input contains the k closest training instances in the feature space. The productivity will depend on whether k-NN is employed for classification or regression.

2.1.3 Pixel Cluster Detection Technique:

Binny Thakral and Manoj Kumar[7] have developed new strategy for for the segmentation of conjuncts, and overlapping characters in Devanagari script on Hindi language.which gave

them 94.5% accuracy for various input characters. In the proposed framework they have actualized pixel cluster identification technique. It is used for segmentation of touching and overlapping characters. Basically, header line is the line consisting of maximum number of black pixels, so after detecting the maximum black pixels in single horizontal line, convert that pixels into white. Afterwards, discover the crevices or gaps between the characters of the word and store these values into an array. At that point, locate the pixel cluster by getting the mid of all array values. In the event, if cluster is distinguished in the characters, then find the midpoint of gathering of the pixels which structure the cluster. Store the midpoint and after that segment the word from computed midpoints. The point where cluster is detected, just segment the characters by adding vertical lines in between.

2.2 Feature Extraction:

An image is a matrix consisting of certain pixel values. Feature extraction process helps to reduce this matrix size and forms a feature vector which contains information of image like its texture, area, shape, colour, object, edges, etc. The feature extraction used here are discrete wavelet transform (DWT) and moment features extraction.

2.2.1 Zoning Algorithm:

Zoning is another feature extraction technique that is based on dividing the image into zones and then calculate the density of white or black pixels in each zone. In Optical character recognition on heterogeneous SoC for HD automatic number plate recognition system [11] have used zoning as feature extraction technique.

Zoning algorithm is one of popular technique for extracting features in pattern recognition area. Zoning is a method for local information analysis on partitions of a given pattern. Michael Reynaldo Phangtristua, Jeklin Harefa, Dian Felita Tanotoa [13] have shown comparison between SVM and neural network classifier with various feature extraction technique where zoning is one of them. Each of the character images are divided into 16 zones (4 x 4). So, each zone contains 64 x 64 pixel. After the image already divided, the features that exist in each part are computed (get the average pixel density feature using equation (1):

$$d(i) = \frac{x}{n}$$

Eq 2.1: Pixel Density

Where $d(i)$ is the density feature of zone- i , x is the number of foreground pixels and n is total number of pixels. Using zoning for feature extraction and SVM Classifier gives accuracy of 81.21% and with ANN 86.07% for dataset shown in [13].

2.2.2 Projection Profile:

Projection profile is one of feature extraction that accumulates the black pixel along rows and columns in the image. Basically, there are two types of projection profile, which are vertical

and horizontal profiles. Let (NN, MM) denotes a binary image in NN lines and MM columns. For Vertical and Horizontal Profile, the equations are

$$P_v[i] = \sum_{j=1}^M S[i, j]$$

Eq 2.2: For vertical projection and

$$P_h[j] = \sum_{i=1}^N S[i, j]$$

Eq2.3: For horizontal projection

Binny Thakral and Manoj Kumar[7] have used Horizontal and vertical Projection Profile for Overlapping and touching characters.

2.2.3 Histogram of Oriented Gradients:

Histogram of Oriented Gradients (HOG) method is widely used in computer vision and image processing area for localization and detection of objects¹². HOG operates on the gradient magnitude of the image. It works by counting the occurrences of gradient orientations in localized parts of an image. Then, the image is divided into several cells and the histograms of gradient directions are formed for each of these cells (since each cell has a fixed gradient orientation bins). To achieve the better accuracy, the histograms have been normalized based on the contrast by calculating a measure of the intensity across block of images and using this value to normalize all cells within a block. Michael Reynaldo Phangtriastua, Jeklin Harefa, Dian Felita Tanotoa [13] Have shown comparison between SVM and ANN classifier with various feature extraction techniques where HOG is one of the technique where the original size of each images are 128 x 128. Subsequently, every image will be divided into 64 x 64 format cells with the numbers of orientation bins are 9. This approach aim at producing 36 features that can be used to feed into the classifier.

2.2.4 The fuzzy Hough transform:

A. Chaudhary [3] have developed an OCR system for hindi language where the fuzzy Hough transform is used for feature extraction. The fuzzy Hough transform treats image points as fuzzy points. The Hough transform for line detection uses mapping $r = x \cos\theta + y \sin\theta$ which provides three important characteristics of line in an image pattern. The parameters r and θ specify position and orientation of line. The count of (r, θ) accumulator cell used in Hough transform implementation specifies number of black pixels lying on it. With this motivation, a number of fuzzy sets on (r, θ) accumulator cells are defined. The height of each synthesized fuzzy set is chosen to define feature element and set of n such feature elements constitute an n -dimensional feature vector for a character.

2.3 Classification:

2.3.1 SVM:

SVM is a very useful supervised method for classification of patterns and characters in image processing. It is generally defined for 2-class problem, which uses optimal hyper-plane to maximize the distance between closest samples of 2 classes. As such it has been successfully applied in a large range of applications and can be extended for many non-linear variants. The power of SVM here is that characters can be separated using a hyper plane, so it uses kernel trick to obtain the better accuracy. the SVM technique is used to find the minimum distance between upper/right/left parts with their characters, where they were separated during segmentation and shirorekha removal. So, the character forms with upper/right/lefts components are also classified efficiently by the system, for example ‘ॐ’ and ‘ॐ’ can be discriminated.

An efficient Devanagari Character Classification using Support Vector Machine (CC-SVM) [8] method is proposed by Shalini Puri and Satya Prakash Singh , which preprocesses the offline scanned imaged documents, normalizes them, segments them using projection profile, removes top line, obtains Shirorekha-Less (SL) characters, extracts their features, and finally, recognizes the SL characters by using SVM classifier. Such method makes use of a pseudo-thesaurus, which stores a set of pre-defined character classes.

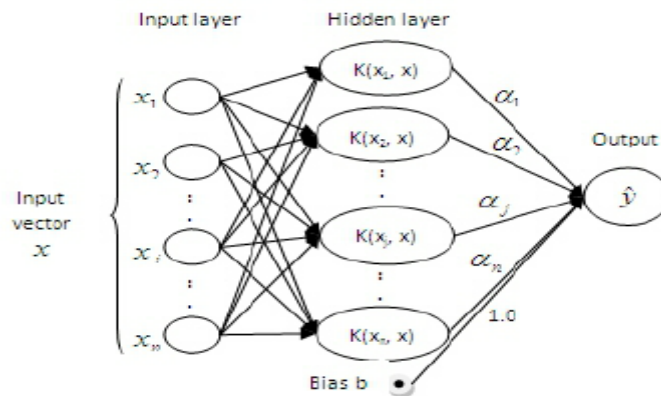


Fig2.1: SVM classifier

2.3.2 ANN :

Artificial Neural Network (ANN) is the algorithm which are designed to resemble the neural network brain system used for pattern recognition based on the input and output data⁶. It is also can be used to find the pattern of data or map the relationship between inputs and outputs.

Michael Reynaldo Phangtrastua, Jeklin Harefa and Dian Felita Tanotoa [13] have used ANN classifier. The algorithm used is Back Propagation Neural Network (BPNN) by reason of it is the simplest model to work and implements. This algorithm generates an appropriate model that can be used to map the output based on the input data. There are three layers of BPNN structure: Input Layer, Hidden Layer and Output and have shown results by comparing various feature extraction techniques with this classifier. Abdul Robby G., Antonia Tandrea, Imelda Susanto, Jeklin Harefa and Andry Chowandaa [12] have developed an OCR for Javanese script using ANN as classifier but using ANN better accuracy is achieved with only single characters

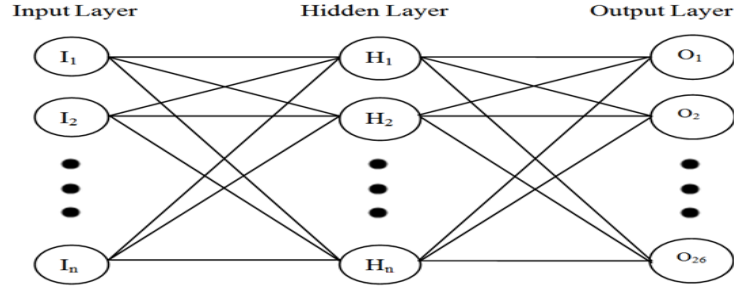


Figure2.2: Architecture of Back Propagation Neural Network

2.3.3 CNN:

It is a class of feedforward neural network with convolutional computation and deep structure. For general large-scale image classification problems, a CNN can be used to construct hierarchical classifiers, and can also be used in fine classification recognition to extract discriminant features of images for other classifiers to learn. An optical character recognition for Sanskrit using Convolution Neural Network [1] was developed by Meduri Avadesh and Navneet Goyal where the Convnets are trained by optimizing on a cross-entropy loss function using mini batch gradient descent and backpropagation which is ideal for digitizing old and poorly maintained material as it is robust to font size and style, image quality and contrast.

It enjoys excellent performance in image classification and has some advantages over traditional technologies like good adaptive performance and high resolution. It integrates the feature extraction function into the multi-layer perceptron via structural reorganization and reducing the weight, and omits the complicated image feature extraction process before recognition. A CNN consists of three main parts. The first part is the input layer. The second part consists of a combination of convolutional layers and pooling layers. The third part consists of fully-connected layers. Haisong Ding, Kai Chen, Qiang Huo have proposed Compressing CNN-DBLSTM models for OCR with teacher-student learning and Tucker decomposition [14]. Based on the architecture of CNN-DBLSTM model, they have proposed an objective function for teacher-student learning that directly matches the feature sequences extracted by CNNs of teacher and student models under the guidance of the succeeding LSTM layers. They use teacher-student learning to transfer the knowledge of a large-size teacher model to a small-size compact student model,

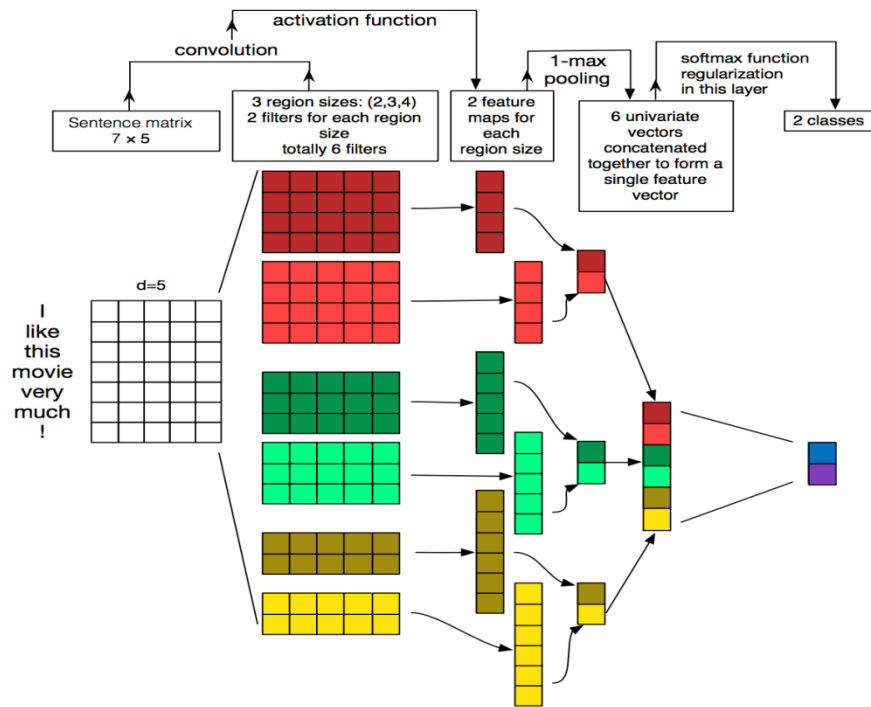
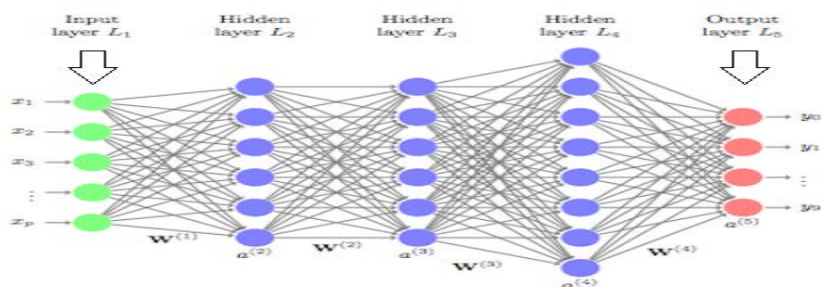


Figure2.3: Convolution Neural network

2.3.4 Deep Learning:

Deep learning has good performance in pattern recognition. Deep learning is therefore considered to be an effective method for handling big data and improving identification performance. Dhara S. Joshi and Yogesh R. Risodkar [6] have developed an OCR for handwritten gujarati characters using Deep learning and have acquired accuracy of about 78.6% but with only single characters. Traditional networks of deep learning that are used to recognize characters have huge network weight and thus their computational burden is very high. To solve these problems, YUE YIN, WEI ZHANG, SHENG HONG¹, JIE YANG, JIAN XIONG, and GUAN GUI [2] Deep Learning-Aided OCR Techniques for Chinese Uppercase Characters have considered Network pruning. In deep networks, the output of most neurons is zero, meaning that most are useless. These inactivated neurons are often redundant and can therefore be removed without affecting the accuracy of the deep network. Re-training after network pruning can achieve the same high accuracy as the original while greatly reducing the network weight.



Figur2.4: Deep neural Network

2.3.5 Hidden Markov Model:

A hidden markov model can be considered a generalization of a mixture model where the hidden variables which control the mixture component to be selected for each observation, are related through a markov process rather than independent of each other . HMM has a property that its state is not defined as a prior information, but are determined automatically based on a database. R. Jayadevan, Satish R. Kolhe, Pradeep M. Patil, and Umapada Pal [10] Have shown comparison of HMM with various other classifiers with an accuracy of 91.3% with 21982 data set(size)

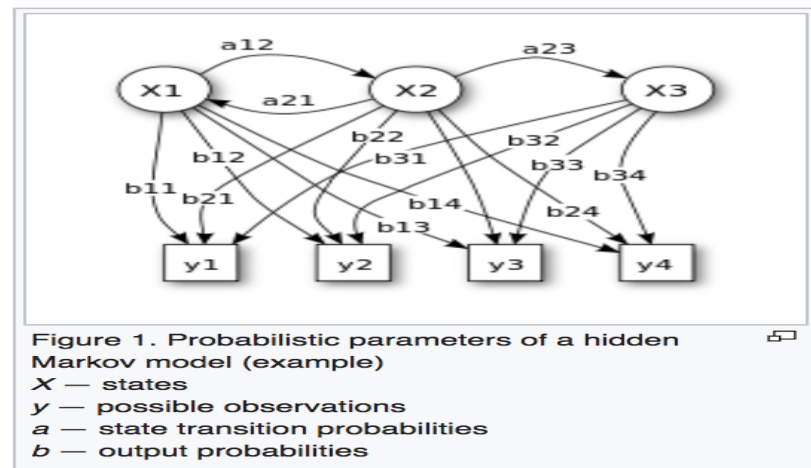


Figure2.5: HMM Classifier

CHAPTER 3

Implementation Platform and Proposed System

There are various platform and various programming languages which can be used for implementation of an OCR system. For the development of the proposed OCR for Sanskrit documents Anaconda navigator is used for python programming. Python is an interpreted, high-level, general-purpose programming language. Created by Guido van Rossum and first released in 1991, Because Python is open and free, it is very easy for other parties to design packages or other software tools that extend Python.

3.1 Implementation Platform

3.1.1 Anaconda Navigator.

Anaconda Navigator is a desktop graphical user interface (GUI) included in Anaconda distribution that allows you to launch applications and easily manage conda packages, environments, and channels without using command-line commands. Navigator can search for packages on Anaconda Cloud or in a local Anaconda Repository. It is available for Windows, macOS, and Linux.

3.1.2 Spyder:

Spyder is a powerful scientific environment written in Python, for Python, and designed by and for scientists, engineers and data analysts. It offers a unique combination of the advanced editing, analysis, debugging, and profiling functionality of a comprehensive development tool with the data exploration, interactive execution, deep inspection, and beautiful visualization capabilities of a scientific package.

3.2 Proposed System

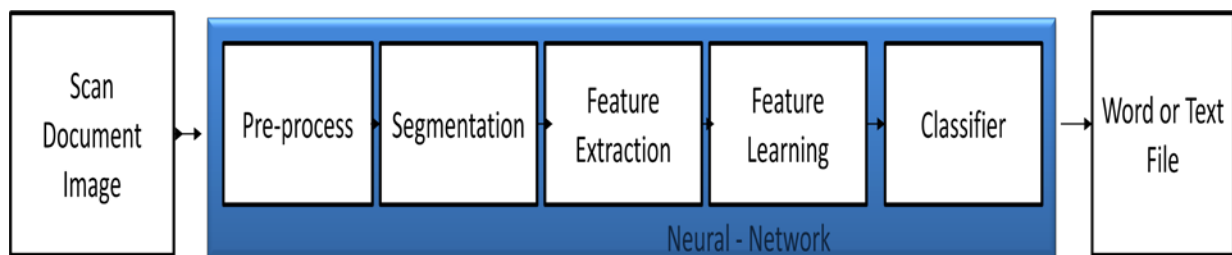


Figure 3.1: Proposed System Diagram

3.2.1 Dataset:

For our proposed system Dataset which consists of total 12912 images comprising of 2880 images of numbers, 2652 images of vowels, 7380 images of consonants of Sanskrit characters is being collected which would be used for training the neural network.



Figure 3.2: Sample dataset of Sanskrit characters

3.2.2 Pre-Processing:

Pre-Processing the image removes noise from the image. In order to achieve higher recognition rates, it is essential to have an effective preprocessing stage.

Preprocessing is performed on scanned by different filters like:

a) Average filter (It takes averages of all pixel values under kernel area)

Average (or mean) filtering is a method of 'smoothing' images by reducing the amount of intensity variation between neighbouring pixels. The average filter works by moving through the image pixel by pixel, replacing each value with the average value of neighbouring pixels, including itself. A major use of averaging filters is in the reduction of “Irrelevant” detail in an image. A spatial averaging filter in which all coefficients are equal is sometimes called a box filter. Also known as low pass filter.

Consider the following 3 by 3 average filter:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

It can be written mathematically as:

$$I_{new}(x, y) = \sum_{j=-1}^1 \sum_{i=-1}^1 1 \times I_{old}(x+i, y+j)$$

$$I_{new_normalized}(x, y) = \frac{1}{\sum_{j=-1}^1 \sum_{i=-1}^1 1} \sum_{j=-1}^1 \sum_{i=-1}^1 1 \times I_{old}(x+i, y+j)$$

Eq3.1: Average filter

b) Gaussian filter (remove Gaussian noise by using Gaussian kernel)

Gaussian Smoothing is the result of blurring an image by a Gaussian function. It is also known as Gaussian blur. Gaussian smoothing is a special case of weighted smoothing, where the coefficients of the smoothing kernel are derived from a Gaussian distribution. The 2D Gaussian smoothing filter is given by the equation shown below

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

Eq 3.2: Gaussian filter

where σ is the variance of the mask

c) Median filter (It takes medians of all pixel values under kernel area)

A filtering window of 3x3 is applied on the image where the central pixel value is replaced with the median value of the corresponding pixels which helps to remove noisy pixel of irrelevant value. It is more useful for removal of salt and pepper type of noise.

252	255	252
252	2	255
255	255	253

Neighbourhood value:

2,252,252,252,**253**,255,255,255,255

Median value: 253

Here the value of 2 is replaced with the value 253 which removes noise and fills the faded portion in the image. After applying the median filter to the original cropped image the output obtained is shown below.

d) Bilateral filter (it reduces noise and preserves the edges, but slower operation)

A bilateral filter is a non-linear, edge-preserving, and noise-reducing smoothing filter for images. It replaces the intensity of each pixel with a weighted average of intensity values from nearby pixels. This weight can be based on a Gaussian distribution. Crucially, the weights depend not only on Euclidean distance of pixels, but also on the radiometric differences (e.g., range differences, such as color intensity, depth distance, etc.). This preserves sharp edges.

The bilateral filter is defined as^{[1][2]}

$$I^{\text{filtered}}(x) = \frac{1}{W_p} \sum_{x_i \in \Omega} I(x_i) f_r(\|I(x_i) - I(x)\|) g_s(\|x_i - x\|),$$

and normalization term, W_p , is defined as

$$W_p = \sum_{x_i \in \Omega} f_r(\|I(x_i) - I(x)\|) g_s(\|x_i - x\|)$$

where

I^{filtered} is the filtered image;

I is the original input image to be filtered;

x are the coordinates of the current pixel to be filtered;

Ω is the window centered in x , so $x_i \in \Omega$ is another pixel;

f_r is the range kernel for smoothing differences in intensities (this function can be a [Gaussian function](#));

g_s is the spatial (or domain) kernel for smoothing differences in coordinates (this function can be a [Gaussian function](#)).

Eq 3.3: Bilateral filter

3.2.3 Segmentation:

Segmentation of an image means to divide it into different group of interest, for example background and foreground. Each pixel in image will be assigned a label which represents a property.

Thresholding is one type of segmentation used by researchers to separate the text from its background it separates the image into pixels only labeled with two different colours.

How It Works

The input to a thresholding operation is typically a grayscale or color image. In the simplest implementation, the output is a binary image representing the segmentation. Black pixels correspond to background and white pixels correspond to foreground (or *vice versa*). In simple implementations, the segmentation is determined by a single parameter known as the *intensity threshold*. In a single pass, each pixel in the image is compared with this threshold. If the pixel's intensity is higher than the threshold, the pixel is set to, say, white in the output. If it is less than the threshold, it is set to black.

Threshold technique is one of the important techniques in image segmentation. This technique can be expressed as:

$$T=T[x, y, p(x, y), f(x, y)]$$

Where T is the threshold value. x, y are the coordinates of the threshold value point. p(x,y), f(x,y) are points the gray level image pixels [9]. Threshold image g(x,y) can be define:

$$g(x,y)=\begin{cases} 1 & \text{if } f(x,y) > T \\ 0 & \text{if } f(x,y) \leq T \end{cases}$$

Eq 3.4: Thresholding Equation

In the proposed system Thresholding is used as segmentation to separate letters from background. Whose output is shown in the results below

3.2.4 Feature Extraction:

Extracting feature helps us to obtain the features which would be useful to train the classifier for example for detecting edge feature kernel which detects edge moves over image matrices and performs convolution the final output obtained is edge detected feature matrices.

Discrete Wavelet Transform:

Using Discrete Wavelet Transform information can be extracted in time domain and frequency domain as well. Hence we can say that Wavelet has a multi-goals properties by deteriorating the sign in various scales.

Mathematically DWT is calculated using the following equation:

$$f(x) = \sum_k c_{j_0}(k) \phi_{j_0,k}(x) + \sum_{j=j_0}^{\infty} \sum_k d_j(k) \psi_{j,k}(x)$$

where j_0 is an arbitrary starting scale

$$c_{j_0}(k) = \langle f(x), \tilde{\phi}_{j_0,k}(x) \rangle = \int f(x) \tilde{\phi}_{j_0,k}(x) dx$$

called the approximation or scaling coefficients

$$d_j(k) = \langle f(x), \tilde{\psi}_{j,k}(x) \rangle = \int f(x) \tilde{\psi}_{j,k}(x) dx$$

called the detail or wavelet coefficients

Eq 3.5: DWT Equation

In image processing the block diagram shown below shows that how DWT works for extracting features from image.

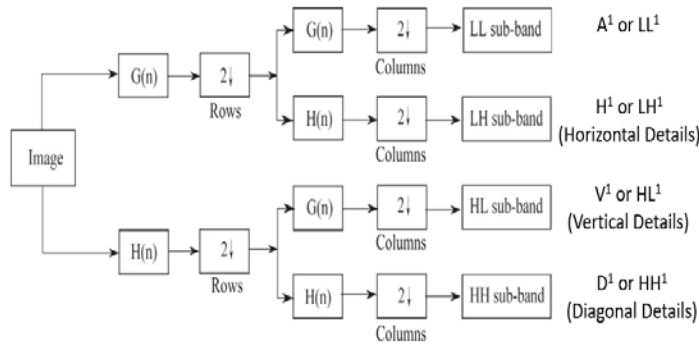


Figure 3.3: Block diagram of DWT

Using this Wavelet feature extraction technique vertical and diagonal features of the image are derived in the form of feature vector which is a row vector which is described as CA and CD respectively as vertical and diagonal feature vectors.

Moment Feature:

To find features like area, perimeter, centroid contour features are used. To find this various features image moment is supposed to be calculated this moment is used to calculate various datas like area, perimeter centroid etc where centroid is given by relation

$$Cx = M10/M00 \text{ and } Cy = M01/M00.$$

Contour Area: It is defined as total number of pixels in the image.

Contour Perimeter: It gives total number of pixels in the boundary. It is also called arc length. argument specify whether shape is a closed contour (if passed True), or just a curve.

2.2.5 Classifiers:

Feature matrix obtained are used to train classifier and a classifier model is generated later the same model is used to detect a new image. In case of an OCR the feature matrix of new

image letter is compared with the image matrix of every letter and by comparison it recognises that the new image is of which letter. Below are classifiers which are implemented.

SVM:

Support vector machine (SVM) is one of the machine learning approach which is used for classification problem. The algorithm used here is simple it creates a line or hyperplane which separates the data into classes. Support vector machine (SVM) used in this experiment is a linear SVM. Since SVM is a machine learning approach method to design a classifier feature vectors are to be extracted by the designer and then given as input to the SVM model for training. First SVM is trained with wavelet feature matrix and then combination of wavelet and moment feature is done.

CNN:

Convolution neural network is a deep learning approach to design a classifier. A CNN is combination of different layers which include convolution layer, maxpooling layer, dropout layer and dense layer the proposed classifier model designed is shown blow. In the proposed classifier model a CNN classifier which possess two deep layers have been designed. Here of 12911 image 80% images are used for training the neural network and rest 20% for testing of the data. Before providing input to the CNN images are first resized from 28x28x3 to 32x32x1 and converted from RGB to grey and certain morphological operations are performed which includes filtering for which median filter is used and segmentation of letters from background which is done by performing thresholding operation on the image. The dataset used is divided into 58 different classes each class of single letter and numeric value consisting of more than 200 images which will be used as input in the neural network.

Model: "sequential_4"

Layer (type)	Output Shape	Param #
conv2d_7 (Conv2D)	(None, 27, 27, 64)	320
max_pooling2d_7 (MaxPooling2)	(None, 13, 13, 64)	0
dropout_10 (Dropout)	(None, 13, 13, 64)	0
conv2d_8 (Conv2D)	(None, 12, 12, 32)	8224
max_pooling2d_8 (MaxPooling2)	(None, 6, 6, 32)	0
dropout_11 (Dropout)	(None, 6, 6, 32)	0
flatten_4 (Flatten)	(None, 1152)	0
dense_7 (Dense)	(None, 128)	147584
dropout_12 (Dropout)	(None, 128)	0
dense_8 (Dense)	(None, 58)	7482
Total params: 163,610		
Trainable params: 163,610		
Non-trainable params: 0		

Figure 3.4: Proposed CNN layers

In the designed layer Activation function used is relu and kernel of 2x2 is used on 28x28x1 image and convolution operation is performed between the image matrix and the kernel matrix. 'Relu' function is used to provide non-linearity .After that maxpooling operation performed by 2x2 matrix reduces the image matrix by half i.e. matrices of 28x28 is reduced to 13x13 and considers only essential information. The same operation is performed by the second layer of the CNN. Dropout layer used after that would deactivate around 25% of neurons that give same output to avoid the problem of overfitting. After that the feature matrices obtained is flatten which converts feature map to a single column and passed to fully connected layer which is formed by two dense layers. Here dense layers connects. each output layer with input layer thus forming a fully connected network each layer with that classifies and Recognize image from 58 classes.

CHAPTER 4

Results

An OCR System consists of the following steps, starting with Pre-Processing, segmentation, Feature extraction and classification. The results shown below shows the output of various filters which can be used in Preprocessing for noise removal and after preprocessing output of thresholding segmentation is shown using these outputs comparison can be done that which filter gives better result for removing noise after that machine learning and deep learning approaches are used to design a classifier and the better algorithm is used for designing the OCR.

4.1 Preprocessing on grey scale image of sanskrit document :

System input

Input to the system is a Sanskrit document the image below shows a part of noisy document

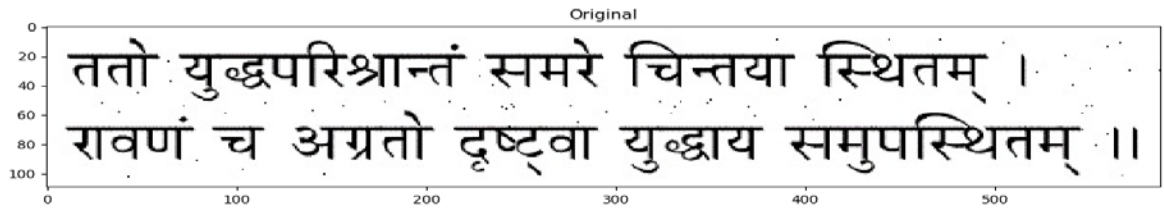


Figure 4.1: Original scanned document

Median filter

The output below is obtained when median filter is used for noise removal .



Figure 4.2 :Applying Median filter on document

Gaussian filter

The output below is obtained when gaussian filter is applied for noise removal .

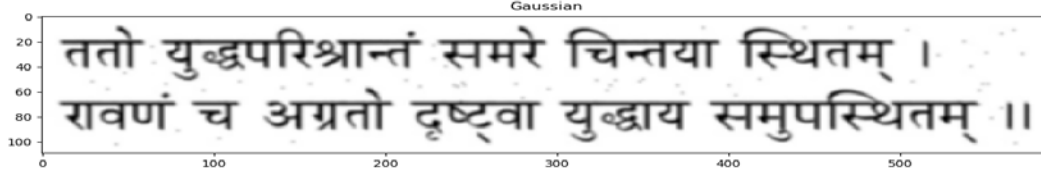


Figure 4.3: Applying Gaussian filter on document

Bilateral filter

The output below is obtained when bilateral filter is applied for noise removal .

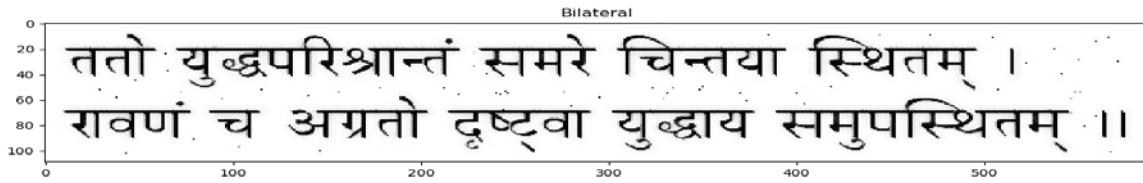


Figure 4.4 : Applying Bilateral filter on document

Average filter

The output below is obtained when average filter is applied for noise removal .

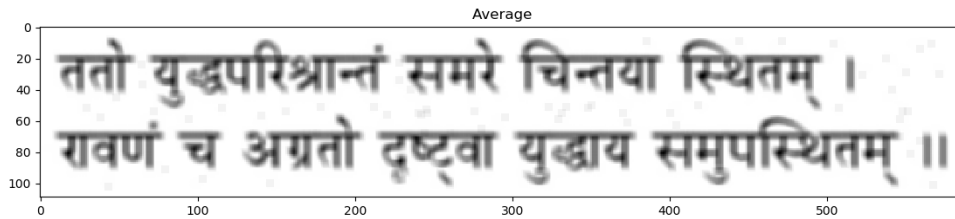


Figure 4.5: Applying Average Filter on document

4.2 Preprocessing on image of colored document :

System input

Input to the system is a Sanskrit document the image below is of a colored document which is given as an input to the system it also contains broken and faded lines and different font size in same document .

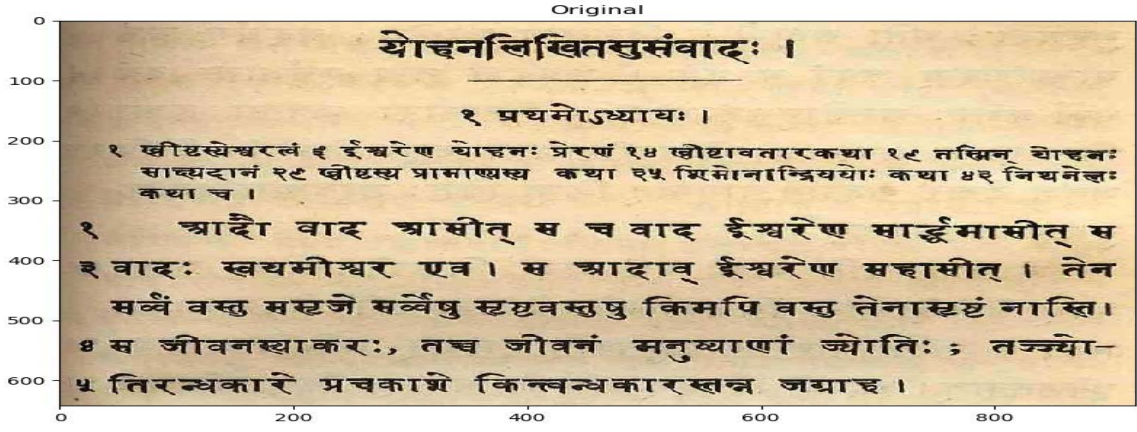


Figure 4.6 :original coloured scanned document

Median filter

The output below is obtained when median filter is applied for noise removal

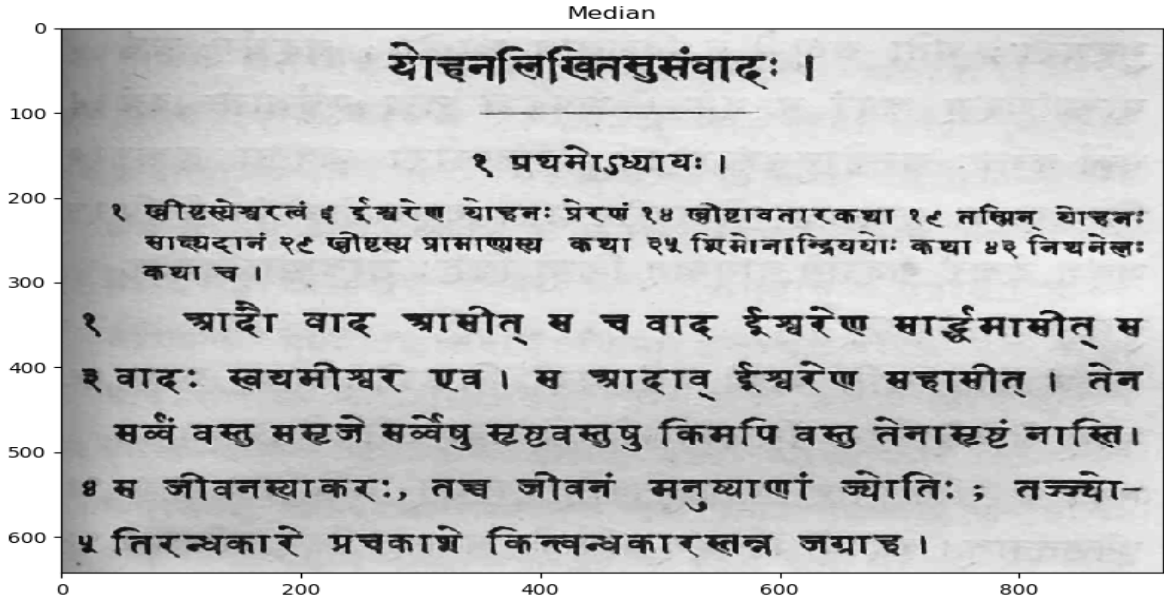


Figure 4.7: Applying Median filter on document

Gaussian filter

The output below is obtained when gaussian filter is applied for noise removal

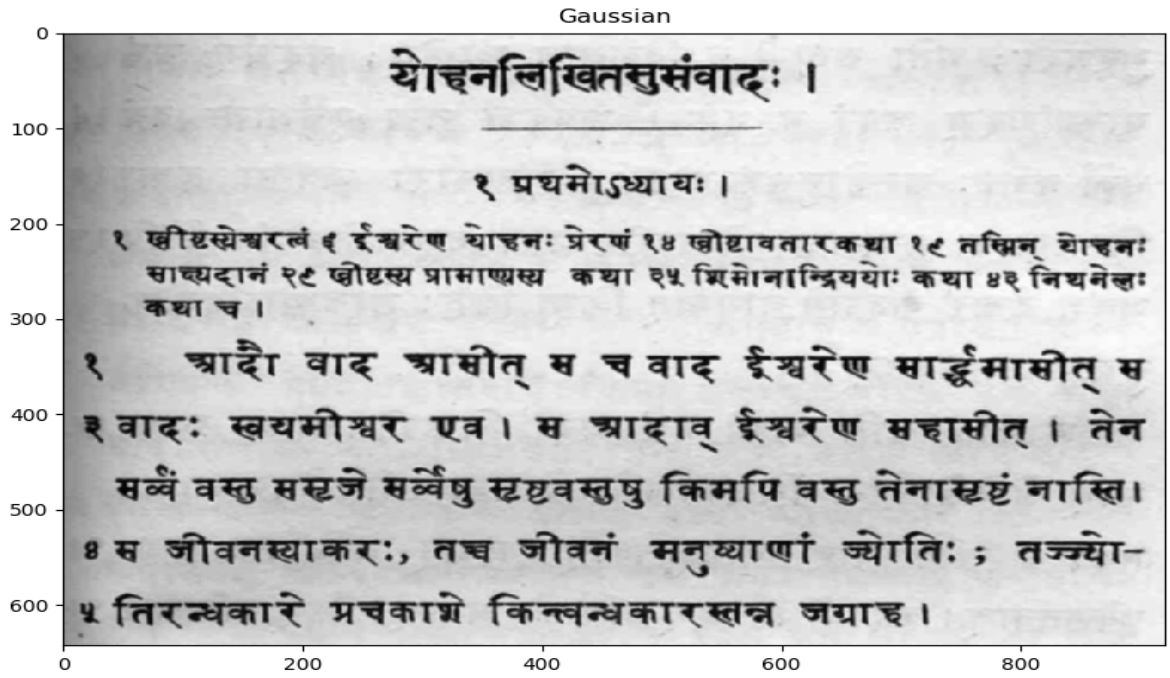


Figure 4.8: Applying Gaussian filter on document

Bilateral filter

The output below is obtained when bilateral filter is applied for noise removal

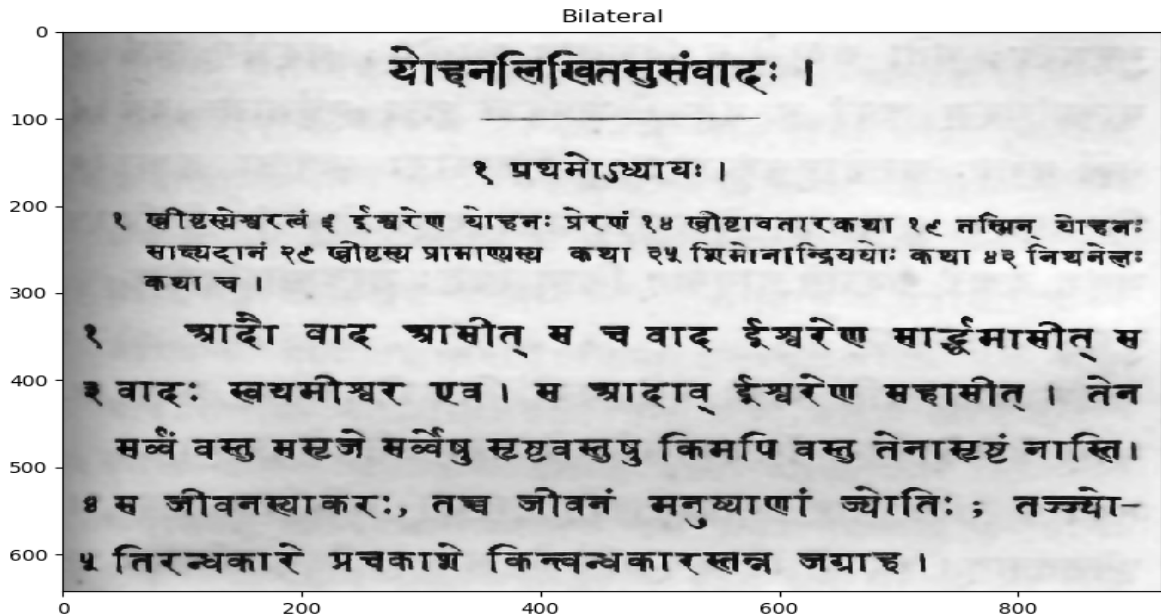


Figure 4.9: Applying Bilateral filter on document

Average filter

The output below is obtained when average filter is applied for noise removal

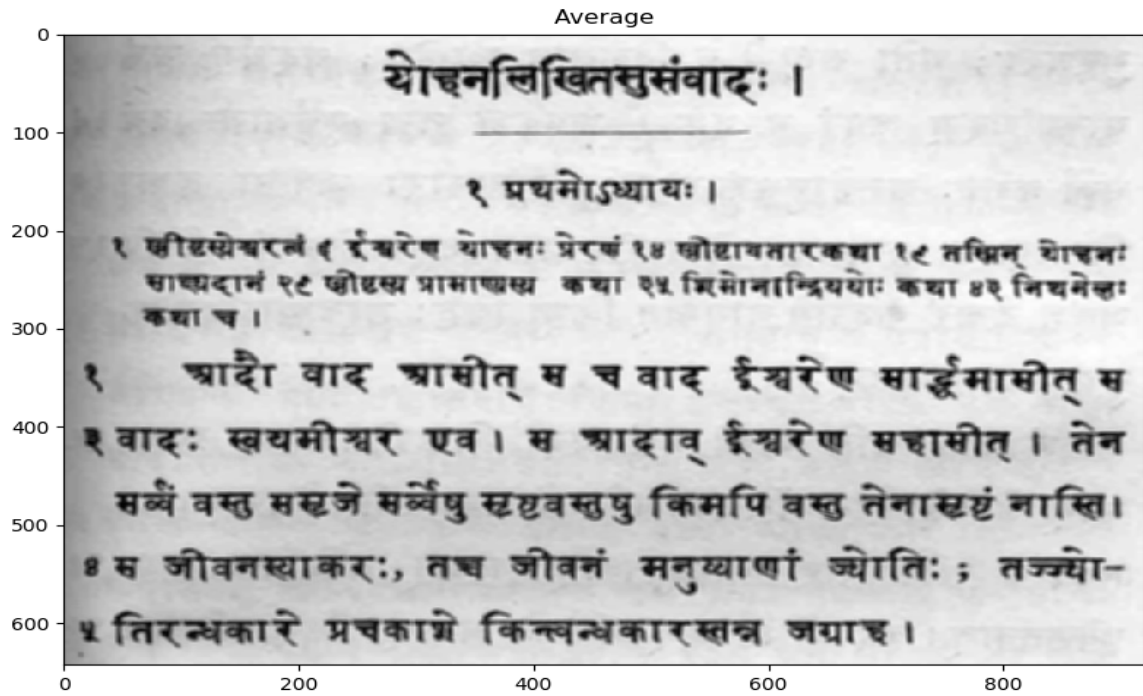


Figure 4.10: Applying Average Filter on document

4.3 Segmentation

Thresholding segmentation

After preprocessing and obtaining a clean document segmentation is performed the image below is the output obtained after performing thresholding segmentation.

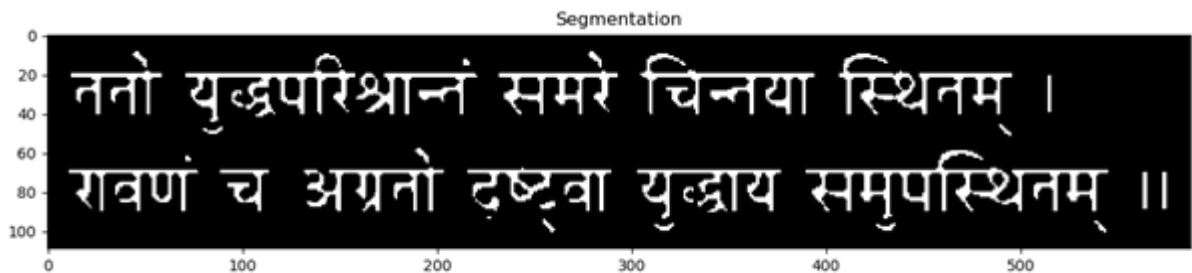


Figure 4.11: Output after segmentation

Thresholding segmentation

After preprocessing and obtaining a clean document segmentation is performed the image below is the output obtained after performing thresholding segmentation on the document which have different font size.

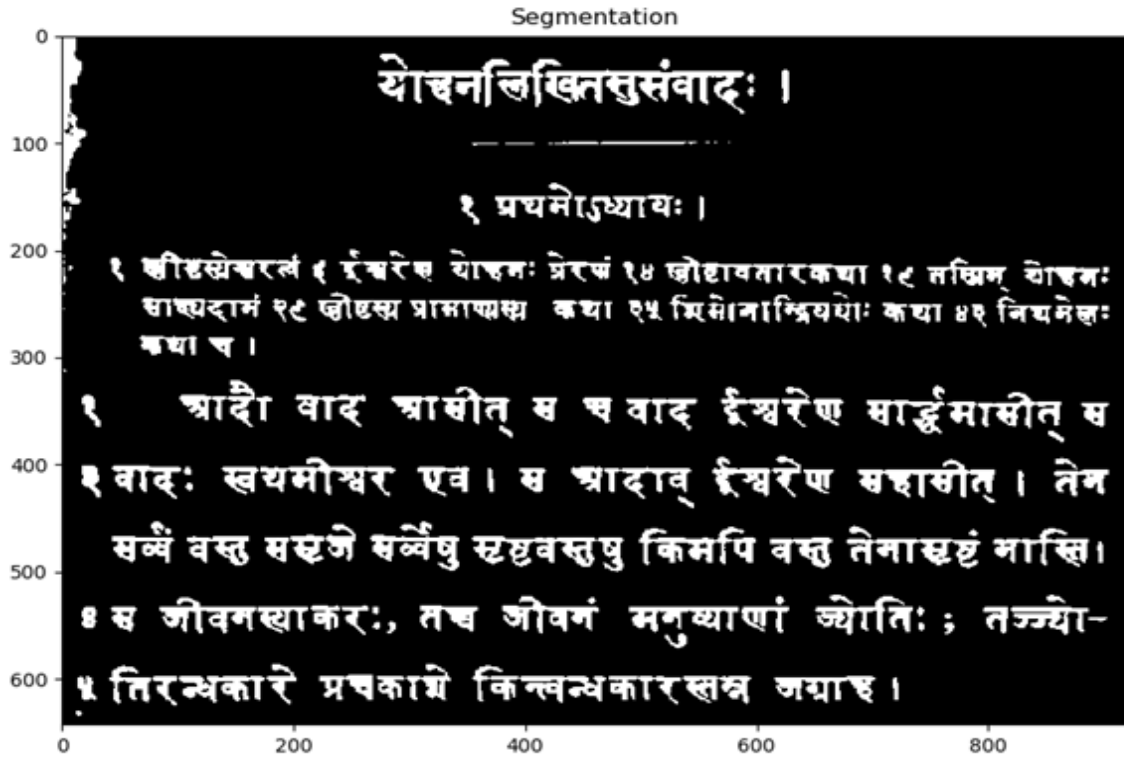


Figure 4.12: Segmentation on document of different font size

4.4 Classifier Output:

MODEL ACCURACY and MODEL LOSS GRAPHS of CNN classifier

The graph below shows model accuracy and model loss graph of train test data of CNN classifier with 20 epoch.

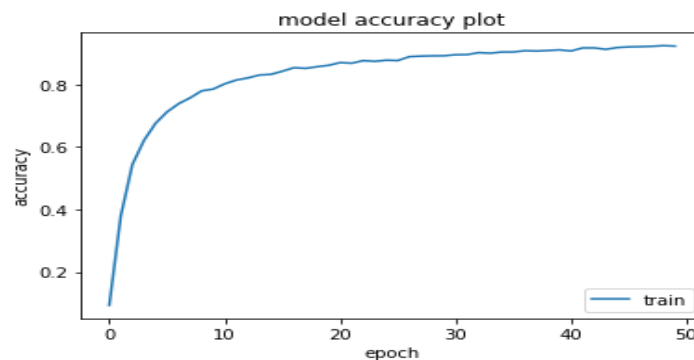


Figure 4.13 : model accuracy

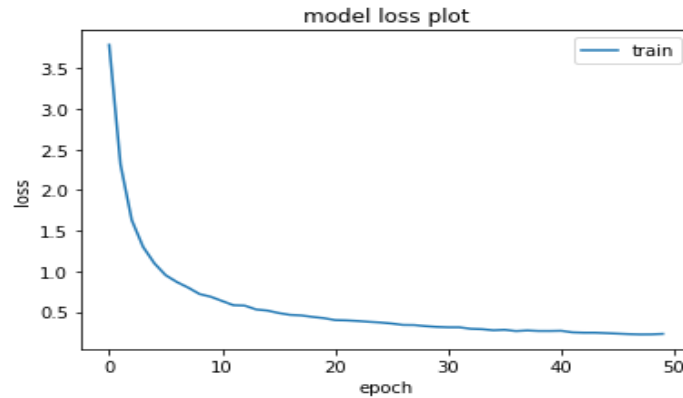


Figure 4.14: model loss graph

Accuracy of Classifier:

Classifier	Time Training	Accuracy
SVM		
Wavelet features	6 min 23 sec	72.164%
Wavelet+Moment features	6 min 46 sec	70.368%
CNN		
Epoch:20	23min 04sec	90.64%
Epoch:50	2 hrs 11min	92.41%

Table: Time required to train and test various classifiers

The table shown above shows that how much time is required to train both the classifier, and how much accuracy each classifier provides, one is designed using machine learning algorithm i.e. SVM classifier where first classifier is trained using only wavelet feature which is combination of diagonal and vertical feature matrix and combination of wavelet and moment . While for deep learning Convolution Neural Network based classifier is designed which takes more time to train feature. The classifiers are trained without GPU on CPU with 64 bit i3 processor and shows that CNN provides more accuracy compared to SVM classifier.

4.5 Final GUI Output:

The following GUI is created using python language in the same spyder software where Classifiers are trained . The GUI output has buttons which include Train, Browse image, Segment, Recognize which recognizes words letters and sentences and write text and at last number entity button to separate out buttons. The outputs are shown in the step by step format:

Clicking on Train button:

When Train button is pressed outputs such as model accuracy plot and model loss plot are printed on the output portion which is generated when we train the classifier.

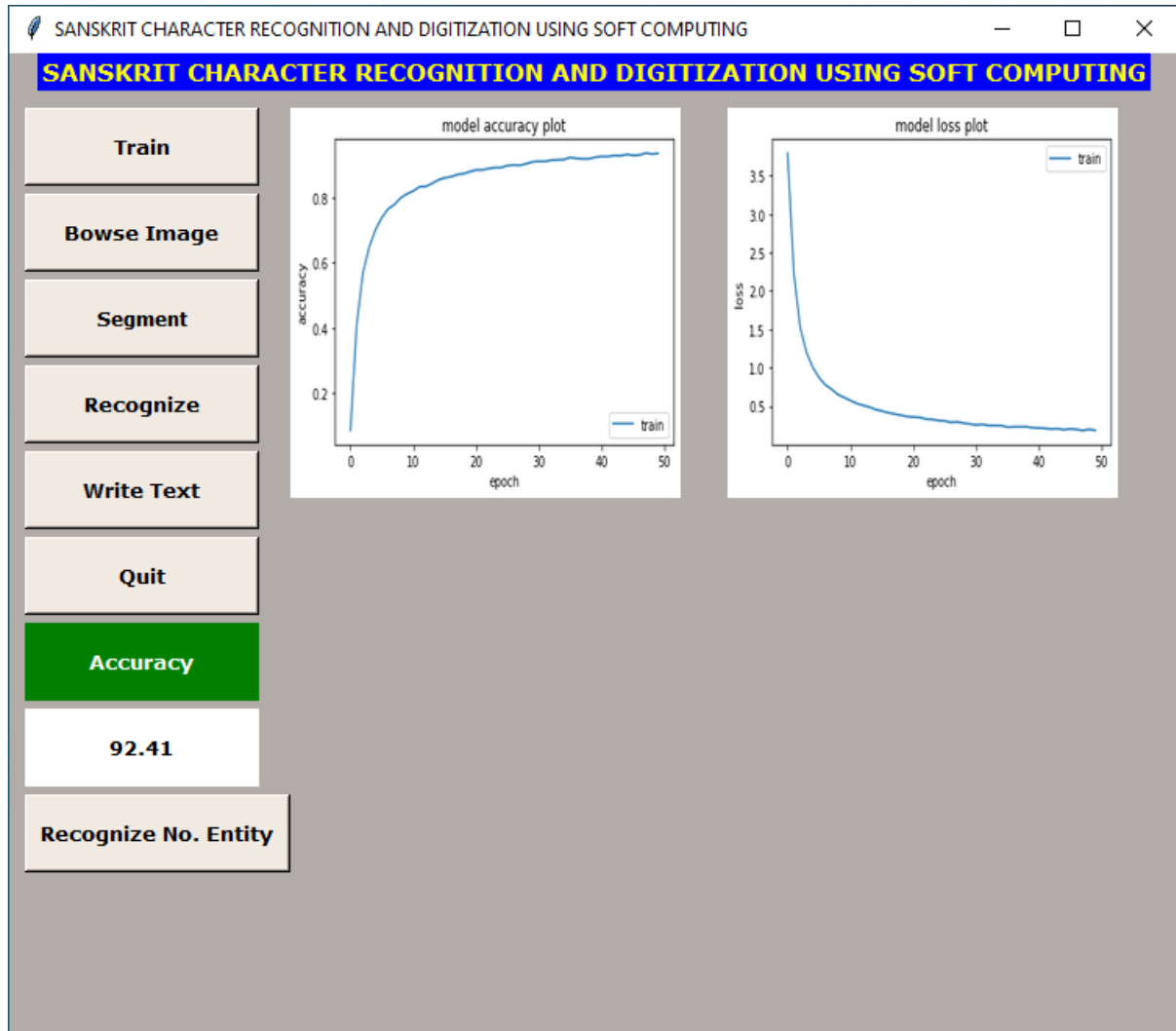


Figure 4.15: Model graphs as output in GUI

Clicking on Browse Image button:

The Browse Image Button allows us to select the image of the document which we want to recognise and digitize .

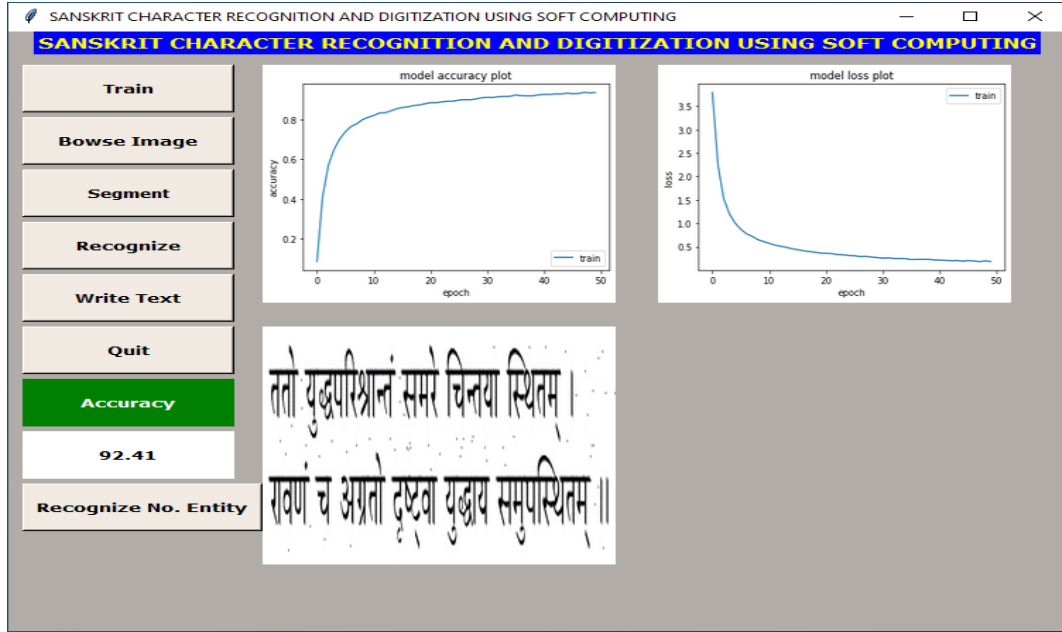


Figure 4.16: Output on clicking Browse Image button

Clicking on Segment Button:

The output obtained when we click on segment button is the segmented and filtered image of the Sanskrit text that was browsed

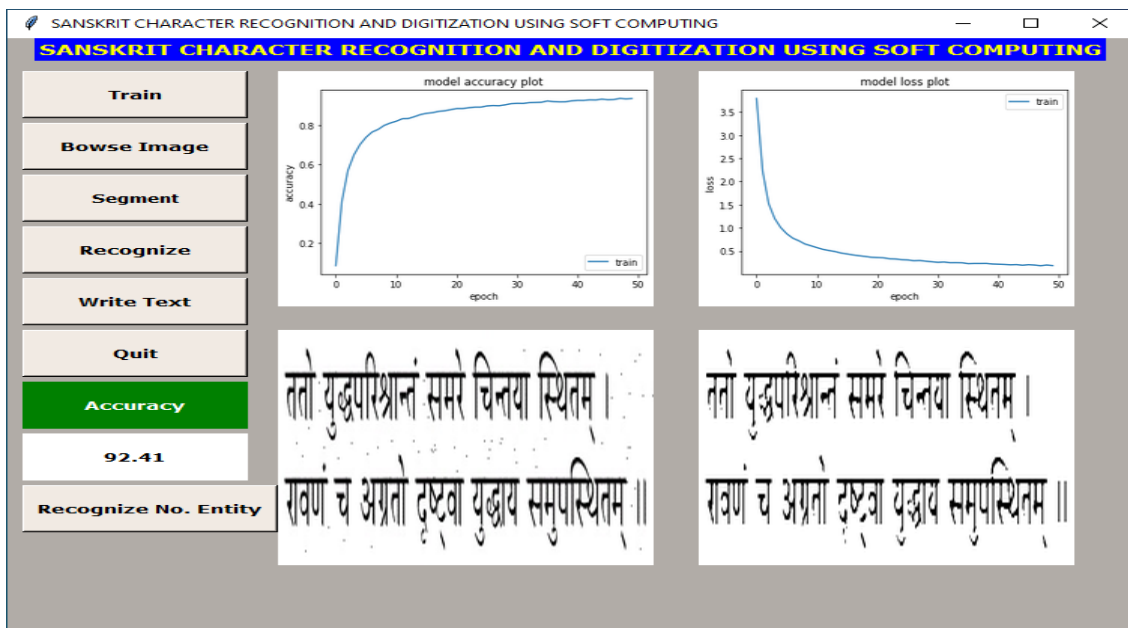


Figure4.17: segmented and filtered image as output

Recognition of each words and sentences:

Clicking on Recognize button helps to separate words from sentences and get it available in editable form.

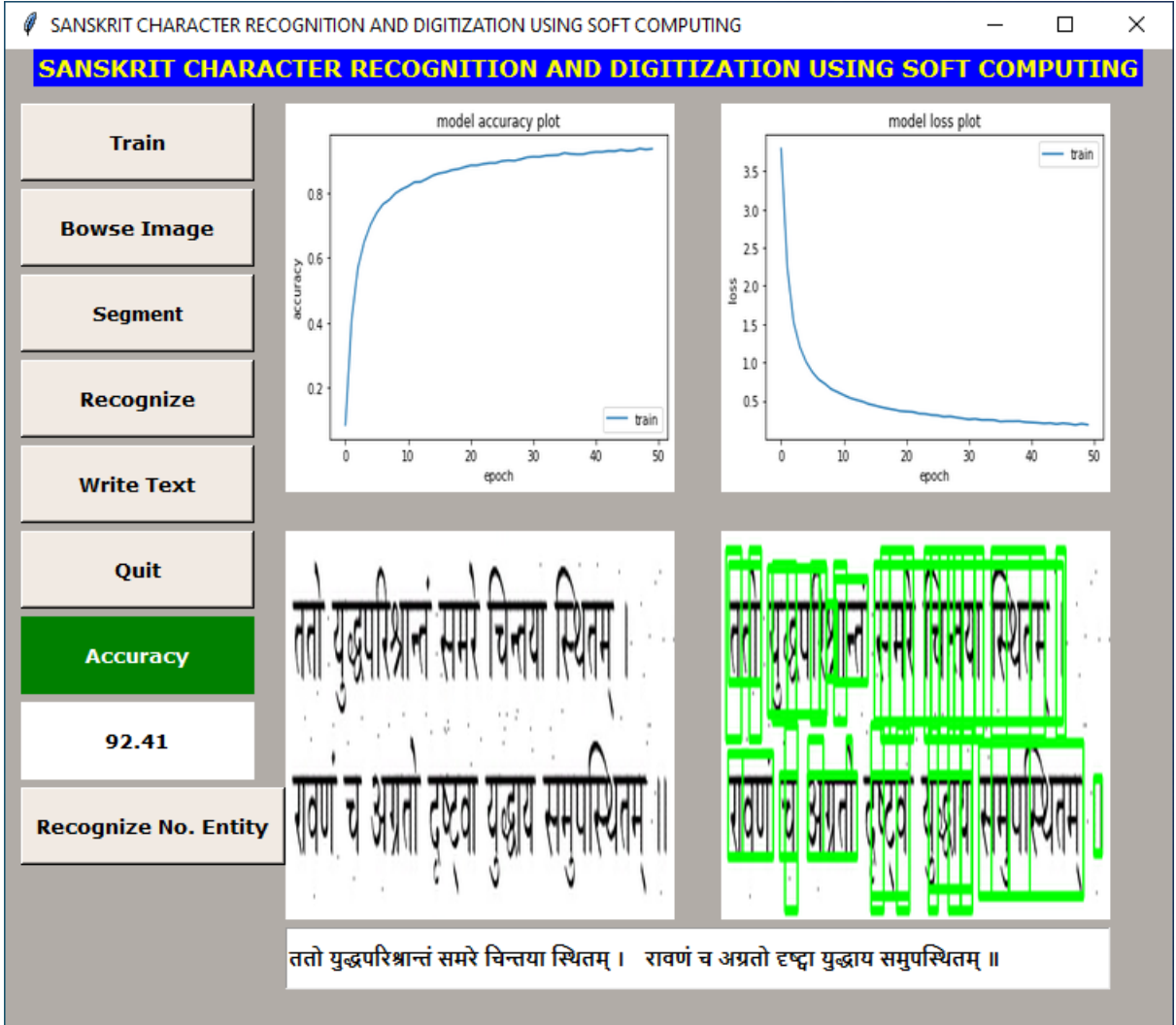


Figure 4.18: Recognizing of letters, words and sentences

It can be clearly seen that the OCR have recognised each word and sentence in correct form as shown in the figure above hence it can be said that recognizing of word and sentences is accurate with proper space between words not too high and not too low

Output obtained after clicking on write button:

Clicking on write button generates a text file where the recognized Sanskrit letters, words and sentences are printed and hence we obtain editable document of scanned image of Sanskrit document

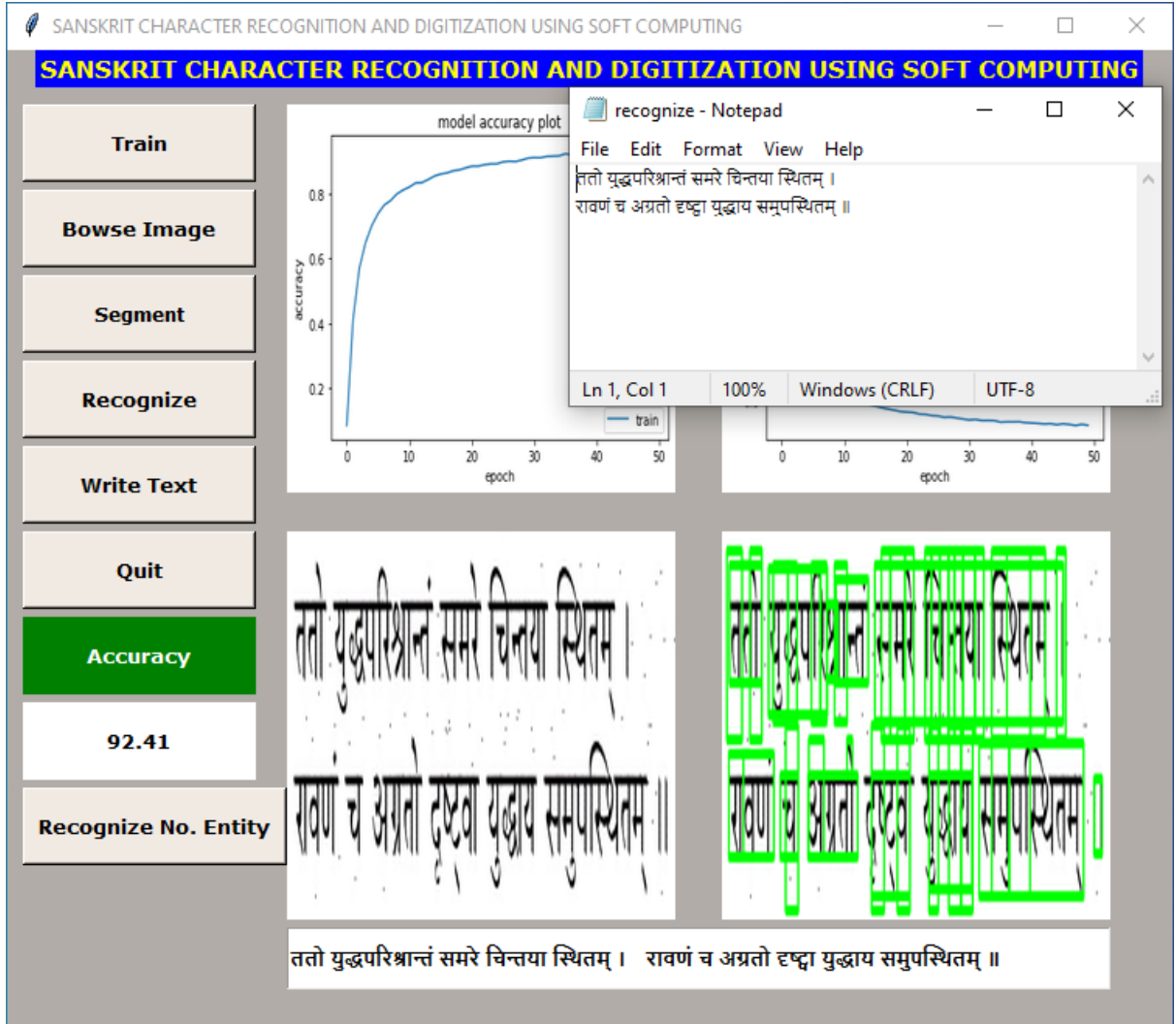


Figure 4.19: Output after clicking Write button

Outputs obtained for coloured background:

The Following OCR works for both documents one with white background and one with coloured background the images shown below shows the output for coloured background Sanskrit document image. With small difference in font size in the same page also the OCR works correctly and gives output with high accuracy. As shown in figure below.

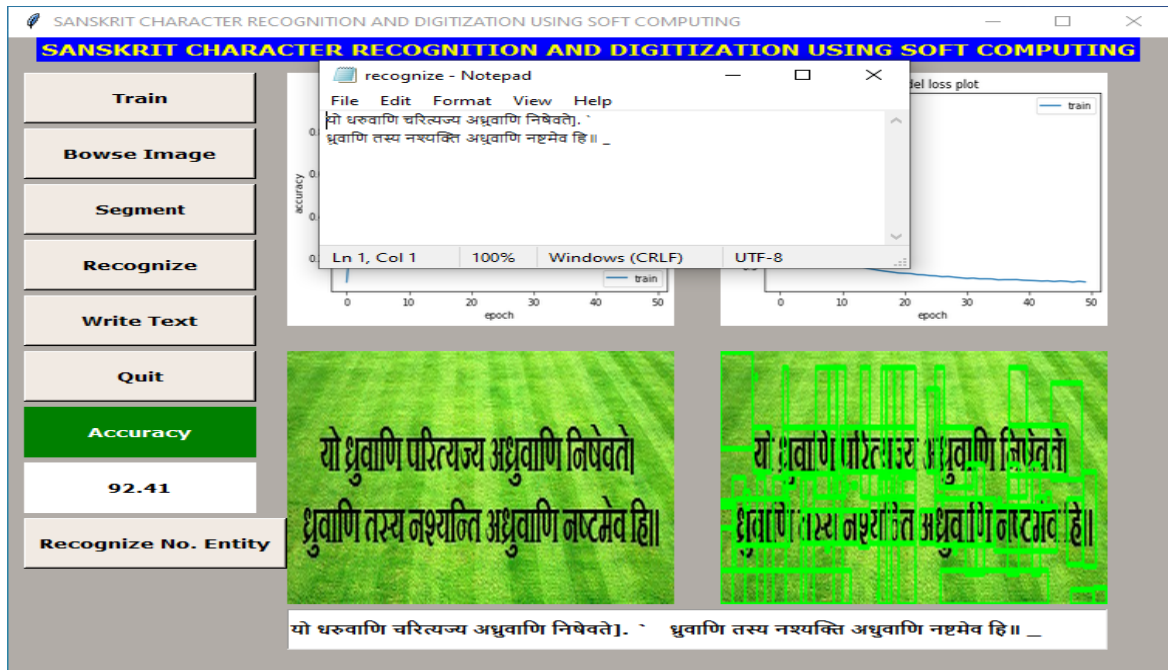
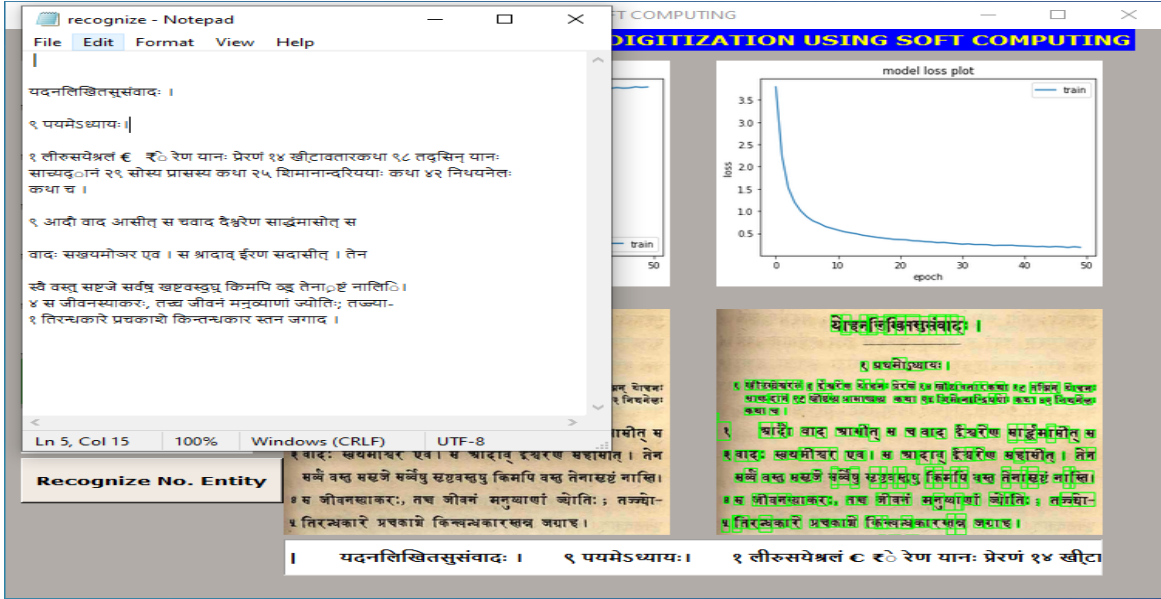


Figure 4.20: Recognizing of Sanskrit words printed on coloured background

Obtaining number entities:

The following OCR also finds numbers in the form of entity. The output is shown as below.

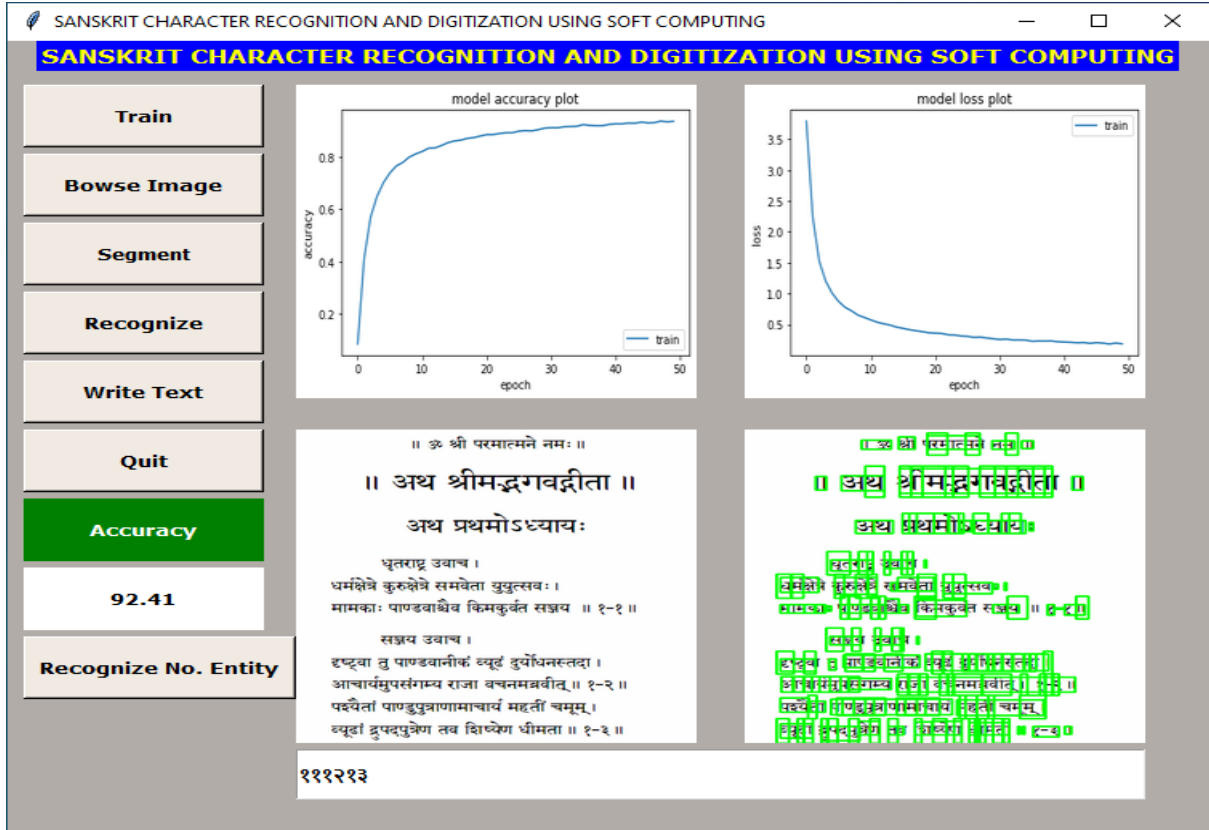


Figure 4.21: Recognizing number Entity

Comparison of existing system with proposed system:

The table shown below shows how the proposed system is different from previous systems

	Existing systems	Proposed systems
Method used	CNN	CNN
Hardware used	GPU	No
Single character	Yes	Yes
Words	No	Yes
Sentences	No	Yes
Entity	No	Yes

Table 4.2 : Existing Vs Proposed

CHAPTER 5

Work Plan

Activity	Months									
	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	April	May
Counselling	√									
Literature Collection	√	√	√	√	√					
Finalizing the Topic	√	√								
Study of Various Preprocessing Technique	√	√	√							
Stimulation in Python			√	√	√					
Writing Dissertation Phase-I Report				√	√					
Study of various Classifier, Keras and Tensorflow Environment in Anaconda						√				
Implementation of Classifier in tensorflow Environment							√			
Creating GUI for OCR								√		
Testing of OCR for various documents									√	
Preparation of thesis								√	√	√

Table 5.1 : Work Plan

CHAPTER 6

Conclusion

Many OCR have been developed but most of them only recognizes the letters or single character of Sanskrit which are provided in cropped form from the document as input to the OCR . Line segmentation in handwritten documents is referenced in previous research[5]. Existing system does not dealing with segmentation of overlapping lines[2]. While other papers does not segmented the touching characters in middle zone and upper zone of printed Sanskrit script and faded character [1]. The work on line segmentation, consonant segmentation, upper modifier Segmentation and lower modifier segmentation and half character segmentation in OCR is not presented yet. So need an to develop an OCR that could accurately recognize and digitize sanskrit documents that are poorly maintained and also have different fonts, have overlapping characters, touching lines and faded letters. As the the main objective is Maintainance of ancient sanskrit documents in digital form before they get completely lost.

The following OCR not only works with recognizing of letters but classifies words and scentences and the scanned Sanskrit document is obtained in the form editable text and provides accuracy upto 92.41%. It Even works with documents with coloured and white background and even if the document have different font size in same page it is able to classify with high accuracuy .The following OCR can also be used for handwritten documents but its accuracy decreases for slanting line and overlapping words.

REFERENCES

Papers

1. Avadesh, M., & Goyal, N. (2018). Optical character recognition for sanskrit using convolution neural networks. *Proceedings - 13th IAPR International Workshop on Document Analysis Systems, DAS 2018*, 447–452. <https://doi.org/10.1109/DAS.2018.50>
2. Ding, H., Chen, K., & Huo, Q. (2019). Compressing CNN-DBLSTM models for OCR with teacher-student learning and Tucker decomposition. *Pattern Recognition*, 96, 106957. <https://doi.org/10.1016/j.patcog.2019.07.002>
3. Phangtriastu, M. R., Harefa, J., & Tanoto, D. F. (2017). Comparison between Neural Network and Support Vector Machine in Optical Character Recognition. *Procedia Computer Science*, 116, 351–357. <https://doi.org/10.1016/j.procs.2017.10.061>
4. Abdul Robby, G., Tandra, A., Susanto, I., Harefa, J., & Chowanda, A. (2019). Implementation of optical character recognition using tesseract with the javanese script target in android application. *Procedia Computer Science*, 157, 499–505. <https://doi.org/10.1016/j.procs.2019.09.006>
5. Farhat, A., Hommos, O., Al-Zawqari, A., Al-Qahtani, A., Bensaali, F., Amira, A., & Zhai, X. (2018). Optical character recognition on heterogeneous SoC for HD automatic number plate recognition system. *Eurasip Journal on Image and Video Processing*, 2018(1). <https://doi.org/10.1186/s13640-018-0298-2>
6. Jayadevan, R., Kolhe, S. R., Patil, P. M., & Pal, U. (2011). Offline recognition of Devanagari script: A survey. *IEEE Transactions on Systems, Man and Cybernetics Part C: Applications and Reviews*, 41(6), 782–796. <https://doi.org/10.1109/TSMCC.2010.2095841>
7. Wei, T. C., Sheikh, U., & Rahman, A. A. H. A. (2018). Improved optical character recognition with deep neural network. *Proceedings - 2018 IEEE 14th International Colloquium on Signal Processing and Its Application, CSPA 2018*, (March), 245–249. <https://doi.org/10.1109/CSPA.2018.8368720>
8. Puri, S., & Singh, S. P. (2019). An efficient Devanagari character classification in printed and handwritten documents using SVM. *Procedia Computer Science*, 152, 111–121. <https://doi.org/10.1016/j.procs.2019.05.033>
9. Thakral, B., & Kumar, M. (2015). Devanagari handwritten text segmentation for overlapping and conjunct characters- A proficient technique. *Proceedings - 2014 3rd International Conference on Reliability, Infocom Technologies and Optimization: Trends and Future Directions, ICRITO 2014*. <https://doi.org/10.1109/ICRITO.2014.7014746>
10. Joshi, D. S., & Risodkar, Y. R. (2018). Deep Learning Based Gujarati Handwritten Character Recognition. *2018 International Conference On Advances in Communication and Computing Technology, ICACCT 2018*, 563–566. <https://doi.org/10.1109/ICACCT.2018.8529410>
11. Bathla, A. K., Gupta, S. K., & Jindal, M. K. (2016). Challenges in Segmentation of Handwritten text for recognition of Devanagari Scripts: A Review. *In the Proceedings of 3rd International Conference on Computing for Sustainable Global Development (INDIACom)*, 2711–2715.
12. Adiga, D., Saluja, R., Agrawal, V., Ramakrishnan, G., Chaudhuri, P., Ramasubramanian, K., & Kulkarni, M. (n.d.). Improving the learnability of classifiers for Sanskrit OCR corrections.

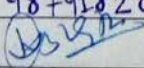
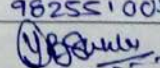
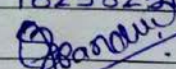
13. Chaudhuri, A., Mandaviya, K., Badelia, P., & K Ghosh, S. (2017). Optical Character Recognition Systems for Different Languages with Soft Computing. *Optical Character Recognition Systems*, 352(January 2018), 9–42. <https://doi.org/10.1007/978-3-319-50252-6>
14. Yin, Y., Zhang, W., Hong, S., Yang, J., Xiong, J., & Gui, G. (2019). Deep Learning-Aided OCR Techniques for Chinese Uppercase Characters in the Application of Internet of Things. *IEEE Access*, 7(c), 47043–47049. <https://doi.org/10.1109/ACCESS.2019.2909401>
15. Nurhayati, Risda, B. C., & Masrurah, S. U. (2014). Optical character recognition feature implementation in cooking recipe application using tesseract Google project. *2014 International Conference on Cyber and IT Service Management, CITSM 2014*, 44–47. <https://doi.org/10.1109/CITSM.2014.7042173>

Patent

1. Felipe Petroski, Raymond Ptucha, Frank Brockler, Paul Hutkowski and Vatsal Singh System and Method of Character Recognition Using Fully Convolution Neural Networks United States Patent Application Publication, US 2018137349A1, May 17, 2018

APPENDIX A

PHOTOCOPY OF REVIEW CARDS

		Gujarat Technological University		Date: <u>17</u> / <u>10</u> / <u>19</u>	
		Master of Engineering (Dissertation Review Card)			
Applicable from 2019 onwards for New Syllabus - effective from Academic year 2018-19.					
Semester:	3	Subject Name & Code:		Internal Review -1 3730002	IR-1
Student's Enrollment:	1 8 0 4 1 0 7 5 4 0 0 3				
Name of Student :	Jadeja Pratikshaba S.				
Student's Mail ID:-	jadejapratiksha08@gmail.com		Branch Code	054	
Students Contact No.	9408845921		Branch Name	Embedded System	
Theme of Title :	Image Processing				
Title of Thesis :	Sanskrit Character Recognition and digitization using soft computing.				
Sr. No.	Comments given by Internal review panel (Please write specific comments)				Modification done based on Comments
1.	Citation Missig. in Introduction				— done —
2.	Modify Objective.				— done —
3.	Include Dataset in PPT.				— done —
4.	Work on Scanned Document.				— done —
5.	Good presentation & work.				— Done —
					<u>Neha</u> Guide's Signature
Note : Above suggestions/modifications/comments must be fulfilled/accommodate by the student in DP 1			Approved ☒ Not Approved ☐	Please tick on any one. If approved, then put marks ≥ 50 %.	
Particulars		Internal Review Panel			
		Expert 1		Expert 2	
Name:	Dr. K R Bhatt		Dr. Y B Shukla		
Institute code & Name:	SVIT (041)		SVIT (041)		
Mobile No.:	9879182051		9825510034		
Signature:					
Particulars		Internal Guide's Details			
		Supervisor		Co-supervisor	
Name:	Dr. Nehal A. Chitelia		Dr. Sandip R Panchal		
Institute code & Name:	SVIT (041)		SVIT-Vasad. (041)		
Mobile No.:	9924873274		9825825079		
Signature:	<u>Neha</u>				

Gujarat Technological University Master of Engineering (Dissertation Review Card)		Date: 13 / 12 / 19	
Applicable from 2019 onwards for New Syllabus - effective from Academic year 2018-19.			
Semester:	3	Subject Name & Code:	Dissertation Phase 1 - 3730003
Student's Enrollment:	180410754003		
Name of Student:	Jadeja Pratikshaba S.		
Student's Mail ID:-	jadejapratiksha08@gmail.com	Branch Code	054
Students Contact No.	9408845921	Branch Name	Embedded System
Theme of Title:	Image Processing		
Title of Thesis:	Sanskrit Character Recognition and digitization using Soft computing		
OBSERVATIONS			
1. Appropriateness of title with proposal. (Yes/ No) <u>Yes</u> 2. Whether the selected theme is appropriate according to the title? (Yes / No) <u>Yes</u> 3. Justify rational of proposed research. (Yes/ No) <u>Yes</u> 4. Clarity of objectives. (Yes/ No) <u>Yes</u>			
Sr. No.	Comments given by External Examiners (DP-I) (Please write specific comments)	Modification done based on Comments	
1	Only Image enhancement applied for preprocted sanskrit. Should work for recognition.	} done Working on it	
2	No Quantified results.		
3	How to decide background?		
4	Digitization may be attempted.		
		 Guide's Signature	
Note : Above suggestions/modifications/comments must be fulfilled/accommodate by the student in IR-2		Approved ☒ Please tick on any one. Not Approved ☐ If approved, then put marks $\geq 50\%$.	
Particulars	Details of External Examiner		
	Expert 1	Expert 2	
Name:	ARUN NANDURBAR KAP	Dr. J. P. Patani	
Institute code & Name:	LDCE A'BAD 028	PRL, A'bad	
Mobile No.: (optional)	9909381501	9426356705	
Email Id:	arunnandurbarkar@ldce.ac.in	jayesh@ppl-res.in	
Signature:			

APPENDIX B

COMPILANCE REPORT OF COMMENTS

Comments given in internal review-1:

Sr No.	Comments	Implementation
1	Citation missing in introduction of ppt	Modified by adding citation in presentation
2	Modify objective	Modified in proper format
3	Include Dataset in ppt	Included Dataset used to train classifier
4	Work on scanned document	Performed Per-processing and segmentation on entire scanned document instead of single word or letter .
5	Good Presentation and Work	-

Comments given in Dp1:

Sr No.	Comments	Implementation
1	Only image enhancement applied should work for recognition	Designed Classifier for recognition of Sanskrit characters
2	No Quantified Results	Included quantified Results in tabular format.
3	How to Decide Background	Recognition is tested for both with white and coloured background but background colour should be light compared to Letter printed
4	Digitization may be attempted	Working on it(Final output obtained would be in computer editable form)

Comments given in Internal Review II:

Sr No.	Comments	Implementation
1	Presentation is good	-
2	Working to Write Paper	Completed
3	Work on Other methods for Comparison	SVM Classifiers was Implemented as well for comparison with CNN.

APPENDIX C

ABBREVIATIONS

OCR : Optical Character Recognition

SVM : Support Vector Machine

CNN : Convolution Neural Network

DWT : Discrete Wavelet Transform

ANN : Artificial Neural Network

GUI : Graphical User Interface

GPU : Graphics Processing Unit

APPENDIX D
PLAGARISM REPORT



Document Information

Analyzed document	THESIS1.docx (D75349467)
Submitted	6/20/2020 9:57:00 AM
Submitted by	S:VIT
Submitter email	mec041owner@gtu.edu.in
Similarity	9%
Analysis address	mec041owner.gtuni@analysis.arkund.com